
REAL-TIME DRONE FPV STREAMING TO VR HEADSET OVER 5G NETWORK

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with the assistance of Gemini 3.1 Pro (Large language model)

Thematic area: 5G / 5G Advanced-based use cases

1 PROBLEM STATEMENT

Usually, First-Person View (FPV) drones need ultra-low latency for control signals and high bandwidth for live video feeds to provide a reliable and deeply engaging experience, especially when the operator is using a VR (Virtual Reality) headset. The deployment of these drones over 5G networks introduces two major practical challenges.

Relying on 5G infrastructure introduces significant communication hurdles for the drone. Because the network dynamically assigns a new IP address upon each reconnection, maintaining a stable, direct link with the ground control station is inherently difficult.

Additionally, routing both bandwidth-intensive video streams and safety-critical MAVLink (Micro Air Vehicle Link) telemetry through a single 5G channel frequently causes network contention. This high-volume video traffic can introduce latency into essential control commands, ultimately compromising the drone's operational reliability during flight.

Although our existing iDronam-based implementation successfully mitigates the dynamic IP addressing challenge, its proprietary nature presents significant drawbacks. Continued reliance on a closed ecosystem inflates long-term operational costs, restricts system flexibility, and limits architectural control. To address these constraints, we are proposing to transition to a completely open-source ground control architecture.

2 PROPOSED SOLUTION

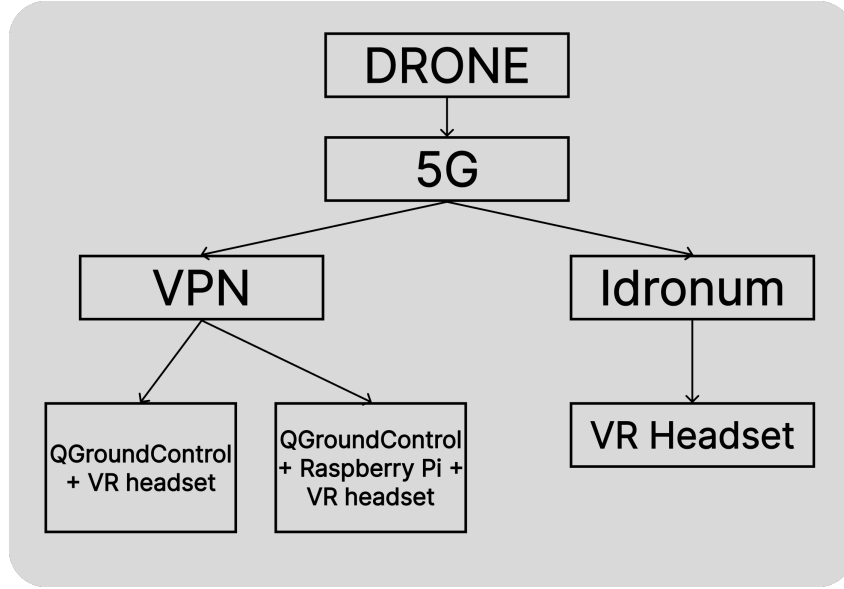


Figure 1: Block diagram of the proposed 5G-enabled FPV drone system showing control and video data paths

We propose the development of a secure, cost-effective, and fully open-source First-Person View (FPV) drone system operating over 5G cellular networks. This architecture integrates a custom drone with virtual reality (VR) headsets to deliver a real-time, highly immersive flight experience. Our current development phase focuses on migrating from the proprietary iDronam platform to a comprehensive open-source ground control stack utilizing QGroundControl (or Mission Planner), MAVLink Router, MediaMTX (an RTSP media server), and WireGuard VPN. Specifically, WireGuard mitigates the dynamic IP limitations of 5G networks by assigning a static virtual IP to the drone, ensuring persistent and stable communication.

This open-source transition guarantees complete infrastructure ownership, transparent data routing, enhanced customizability, reduced operational costs, and strict independence from vendor-specific constraints.

3 TECHNICAL APPROACH AND ARCHITECTURE

Currently, the drone’s native 5G modem manages both high-priority, low-latency MAVLink telemetry and video streaming. At the ground station, QGroundControl or Mission Planner interfaces securely via a WireGuard VPN, while MediaMTX routes the live video directly to a VR headset for an immersive first-person view.

Regarding future enhancements, we propose an interesting architectural enhancement: integrating a dedicated Raspberry Pi equipped with an independent 5G SIM and camera module. This hardware addition physically decouples the communication channels. The drone’s native 5G link will be strictly used for critical MAVLink flight control, maintaining an uninterrupted, low-latency communication. Simultaneously, the Raspberry Pi will independently manage all video capture and transmission using open-source tools like GStreamer and MediaMTX.

This dual-channel approach will make sure that the bandwidth-heavy video congestion or packet loss will never compromise flight control, thereby improving operational safety. Additionally, this lightweight, hardware-agnostic architecture easily supports specialized camera payloads—such as thermal or wide-angle lenses—and is optimized for an easy deployment within any 5G Use Case Lab.

4 USE CASES AND APPLICATIONS

First-Person View (FPV) drones have transitioned from recreational devices to more valuable tools across multiple professional sectors. In media and cinematography, they are used to capture dynamic, high-speed aerial footage. In search and rescue operations, the real-time video feedback allows emergency teams to easily navigate complex environments, such as collapsed structures or dense forests, to locate survivors. Moreover, industrial and agricultural sectors depend on these drones to safely inspect infrastructure such as power lines, telecommunication towers, and bridges without exposing human workers to hazardous conditions. They also provide a safe and secure method for law enforcement and media to monitor large gatherings or disaster zones. In academic settings, FPV systems can also be used as practical platforms for studying aerodynamics, testing control software, and gathering environmental data.

5 EXPECTED IMPACT, SCALABILITY AND MARKET READINESS

This initiative significantly minimizes human risk in hazardous environments, accelerates emergency response protocols, and decreases the operational expenses associated with infrastructure inspection and media production. Moreover, using a fully open-source architecture eliminates recurring proprietary software licenses, allowing for cost-effective, nationwide scalability. This framework is designed to exploit 5G attributes—such as high bandwidth, low latency, and robust mobility—while successfully overcoming practical cellular constraints such as dynamic IP allocation and network contention. In addition, this foundation is highly extensible, and effortlessly supports future advancements including AI-driven autonomous navigation, coordinated drone swarms, and the implementation of 5G network slicing to ensure Quality of Service (QoS).

6 FUTURE ENHANCEMENTS (PRAGATI PHASE)

Here, our main focus will be implementing the dual-channel architecture. We will attach a dedicated Raspberry Pi with its own 5G SIM card and camera module to the drone. This setup will physically separate the communication channels: the drone’s built-in 5G connection will be used only for MAVLink-based flight control and telemetry data (low bandwidth, ultra-low latency), while the Raspberry Pi will independently handle all video capture and streaming using open-source tools such as GStreamer and MediaMTX.

This separation will ensure that any problem with the video stream has zero impact on flight control, significantly improving safety and reliability. It will also give us greater flexibility to use higher-quality or specialised cameras (for example, thermal or wide-angle modules) depending on the mission.

7 CONCLUSION

This project builds on a currently working iDronam-based system and evolves it into a fully open-source, reliable, and scalable 5G-native FPV drone solution that addresses current industry pain points while opening new possibilities across disaster management, infrastructure, agriculture, and creative sectors.